InBody S10

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InBody India

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Introducing the InBody S10

Versatile Body Composition & Body Water Analyzer



Key Features

1) Body Composition Analysis Sheet
2) Body Water Analysis Sheet

Body Composition Analysis Sheet

IB s10 Result Sheet

InBody

ID	Height	Age	Gender	Test Date / Time
	167cm	74	Female	2016.09.19. 10:22

Body Composition Analysis

	Values	Total Body Water	Soft Lean Mass	Fat Free Mass	Weight
Total Body Water (L)	31.6 (29.8~36.4)	31.6			
Protein (kg)	7.7 (8.0~9.8)		39.8 (38.3~46.9)	42.2 (40.6~49.6)	43.5 (49.8~67.4)
Minerals (kg)	2.90 (2.76~3.38)				
Body Fat Mass (kg)	1.3 (11.7~18.8)				

Muscle-Fat Analysis

	Under	Normal	Over
Weight (kg)	43.5		
SMM (kg) Skeletal Muscle Mass		21.2	
Body Fat Mass (kg)	1.3		

Obesity Analysis

	Under	Normal	Over
BMI (kg/m²) Body Mass Index		15.6	
PBF (%) Percent Body Fat	3.0		

Segmental Lean Analysis

Based on ideal weight — Based on current weight

	Under	Normal	Over	ECW Ratio
Right Arm (kg) (%)	2.37	149.0		0.414
Left Arm (kg) (%)	2.40	151.3		0.415
Trunk (kg) (%)	19.1	132.1		0.439
Right Leg (kg) (%)	6.31	125.4		0.443
Left Leg (kg) (%)	6.00	119.4		0.442

ECW Ratio Analysis

	Under	Normal	Over
ECW Ratio			0.437

Body Composition History

Weight (kg)	43.5
SMM (kg) Skeletal Muscle Mass	21.2
PBF (%) Percent Body Fat	3.0
ECW Ratio	0.437

Recent / Total 16.09.19. 10:22

Weight Control

Target Weight	58.6 kg
Weight Control	+ 15.1 kg
Fat Control	+ 12.2 kg
Muscle Control	+ 2.9 kg

Segmental Body Water Analysis

Right Arm	1.86 L	(1.34~2.02)
Left Arm	1.89 L	(1.34~2.02)
Trunk	15.2 L	(13.7~16.8)
Right Leg	5.02 L	(4.78~5.84)
Left Leg	4.77 L	(4.78~5.84)

Water Control

ECW Ratio 0.385	- 2.7 L	/	40.8 kg
ECW Ratio 0.395	- 2.2 L	/	41.3 kg
ECW Ratio 0.405	- 1.7 L	/	41.8 kg

*The water control item shows the water level to be controlled based on the extracellular water ratio. This item shows the water level which varies as the extracellular water ratio is set differently according to the presence or absence of complications as described in a paper published in the 2008 Journal of the Japan Society for Dialysis Therapy (JSDT).

Research Parameters

Intracellular Water	17.8 L	(18.5~22.7)
Extracellular Water	13.8 L	(11.3~13.9)
Basal Metabolic Rate	1281 kcal	(1021~1166)
Waist-Hip Ratio	0.81	(0.75~0.85)
Visceral Fat Area	43.2 cm²	
Bone Mineral Content	2.43 kg	(2.28~2.78)
Body Cell Mass	25.5 kg	(26.5~32.3)
Arm Circumference	24.5 cm	
TBW/FFM	74.8 %	
SMI	6.1 kg/m²	

Reactance

	RA	LA	TR	RL	LL
Xc(Ω) 5kHz	2.4	2.3	0.3	1.4	1.5
50kHz	6.5	7.3	0.3	4.2	3.8
250kHz	10.8	13.7		5.8	5.2

Segmental Phase Angle

1.3°

	RA	LA	TR	RL	LL
φ(°) 50kHz	1.4	1.6	1.1	1.2	1.0

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1kHz	272.9	268.4	14.4	206.9	221.8
5kHz	271.6	267.1	14.3	206.2	221.0
50kHz	266.1	261.8	13.6	202.4	217.0
250kHz	257.7	253.4	13.0	197.0	211.5
500kHz	252.7	248.2	12.6	194.0	208.5
1000kHz	246.5	240.1	12.2	190.2	204.5

[Touch Type , Lying Posture]

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InBody Body Water

ID	Height	Age	Gender	Test Date / Time
	167cm	74	Female	2016.09.19. 10:22

Body Water Composition

	Under	Normal	Over
TBW (L) Total Body Water		31.6	
ICW (L) Intracellular Water		17.8	
ECW (L) Extracellular Water		13.8	

ECW Ratio Analysis

	Under	Normal	Over
ECW Ratio			0.437

Segmental Body Water Analysis

	Under	Normal	Over
Right Arm (L)		1.86	
Left Arm (L)		1.89	
Trunk (L)		15.2	
Right Leg (L)		5.02	
Left Leg (L)		4.77	

Segmental ECW Ratio Analysis

	Right Arm	Left Arm	Trunk	Right Leg	Left Leg
Over					
Slightly Over	0.414	0.415	0.439	0.443	0.442
Normal					

Body Water Composition History

Weight (kg)	43.5
TBW (L) Total Body Water	31.6
ICW (L) Intracellular Water	17.8
ECW (L) Extracellular Water	13.8
ECW Ratio	0.437

Recent / Total 16.09.19. 10:22

Body Composition Analysis

Protein	7.7 kg	(8.0~9.8)
Minerals	2.90 kg	(2.76~3.38)
Body Fat Mass	1.3 kg	(11.7~18.8)
Fat Free Mass	42.2 kg	(40.6~49.6)
Bone Mineral Content	2.43 kg	(2.28~2.78)

Muscle-Fat Analysis

Weight	43.5 kg	(49.8~67.4)
Skeletal Muscle Mass	21.2 kg	(22.3~27.3)
Soft Lean Mass	39.8 kg	(38.3~46.9)
Body Fat Mass	1.3 kg	(11.7~18.8)

Obesity Analysis

BMI	15.6 kg/m²	(18.5~23.0)
PBF	3.0 %	(18.0~28.0)

Water Control

ECW Ratio 0.385	- 2.7 L	/	40.8 kg
ECW Ratio 0.395	- 2.2 L	/	41.3 kg
ECW Ratio 0.405	- 1.7 L	/	41.8 kg

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Research Parameters

Basal Metabolic Rate	1281 kcal	(1021~1166)
Visceral Fat Area	43.2 cm²	
Body Cell Mass	25.5 kg	(26.5~32.3)
Arm Muscle Circumference	24.2 cm	
TBW/FFM	74.8 %	
SMI	6.1 kg/m²	

Reactance

	RA	LA	TR	RL	LL
Xc(Ω) 5kHz	2.4	2.3	0.3	1.4	1.5
50kHz	6.5	7.3	0.3	4.2	3.8
250kHz	10.8	13.7		5.8	5.2

Segmental Phase Angle

1.3°

	RA	LA	TR	RL	LL
φ(°) 50kHz	1.4	1.6	1.1	1.2	1.0

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1kHz	272.9	268.4	14.4	206.9	221.8
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[Touch Type , Lying Posture]

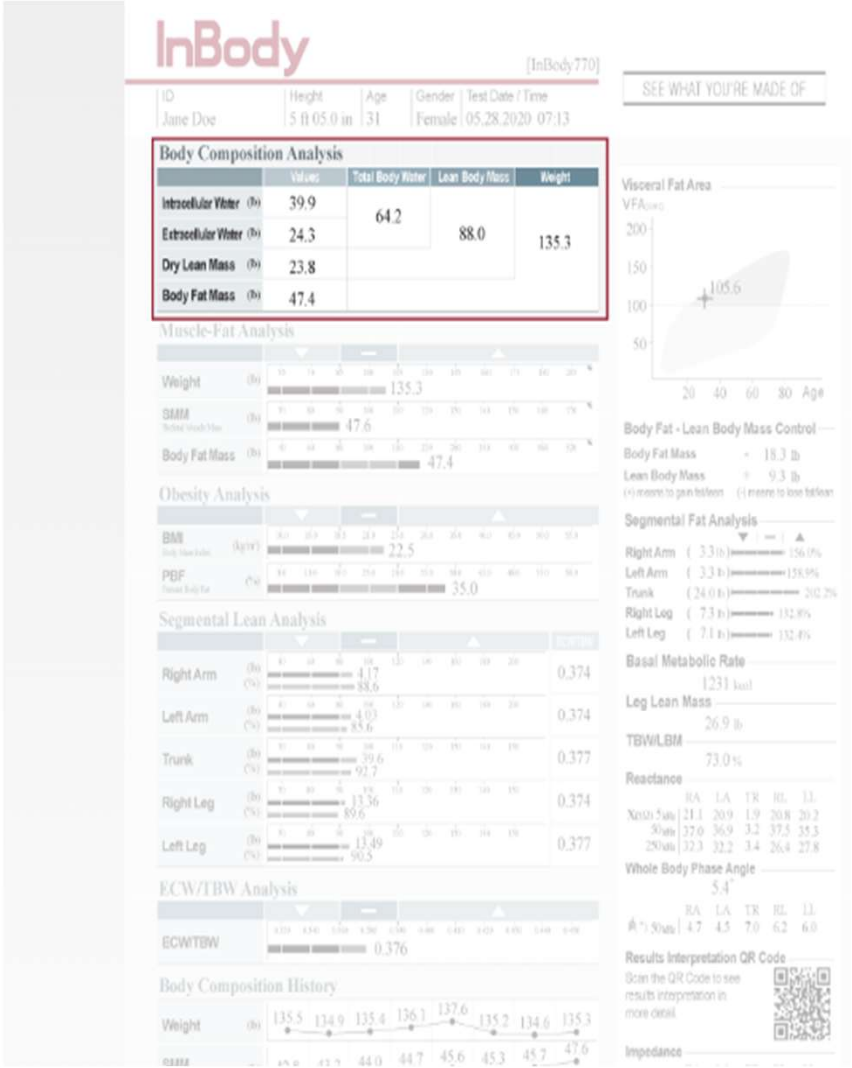
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Chapter 1: Body Composition Analysis & Body Water

Body Composition Analysis

This section gives you a solid breakdown of your overall body composition, displaying these vital measurements in pounds (lb):

- Total Body Water (divided into Intracellular and Extracellular Water)
- Lean Body Mass
- Dry Lean Mass
- Body Fat Mass
- Weight



Chapter 2: Muscle, Fat, & Obesity Risk

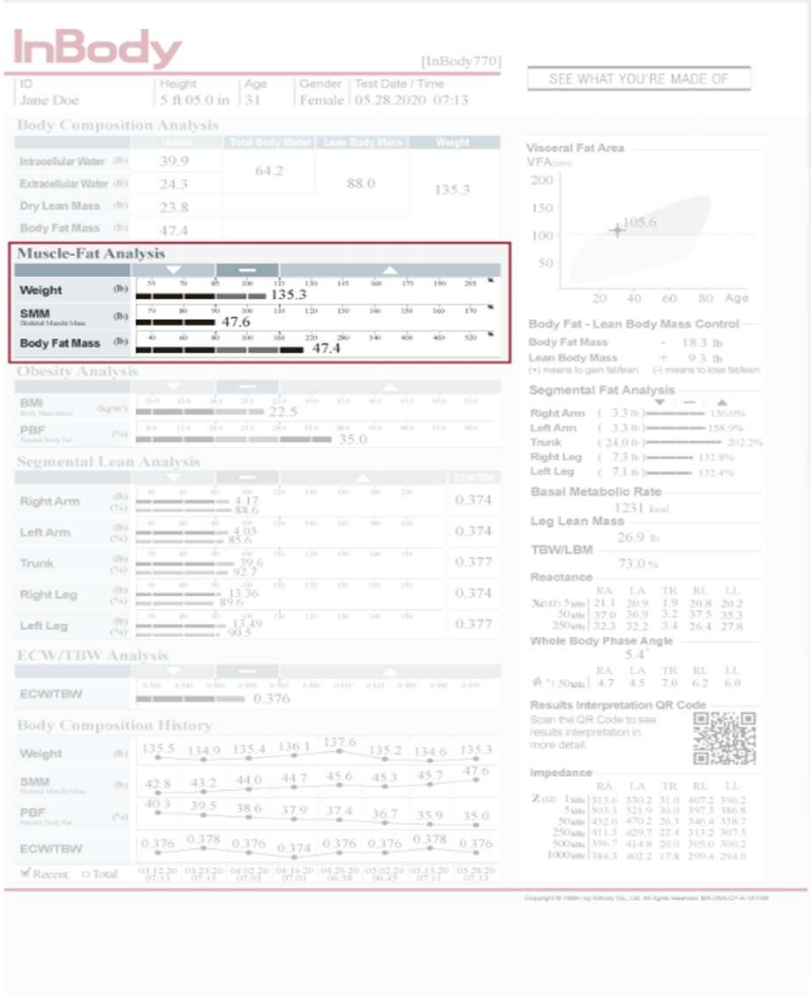
Muscle-Fat Analysis

This section shows how your Weight, Skeletal Muscle Mass, and Body Fat Mass compare to the healthy average range of people of the same height and sex. With Muscle-Fat Analysis, you get a better understanding of where your current body composition is so you can make any changes to get it to where you would like it to be.

Muscle-Fat Analysis is a favorite output for many people because it provides an easy-to-understand overview of body composition and helps narrow the focus of health goals.

Muscle-Fat Analysis has three components:

- **WEIGHT**
  This is your Total Body Weight.
- **SKELETAL MUSCLE MASS (SMM)**
  This is the total weight of your Skeletal Muscle Mass (SMM). SMM is muscle that can be grown and developed through exercise. Unlike Lean Body Mass, which includes everything except body fat, you can confidently interpret an increase in SMM as muscle gain.
- **BODY FAT MASS**
  This is how much body fat you have (both the surface level and internal fat).



Muscle, Fat, & Obesity Risk

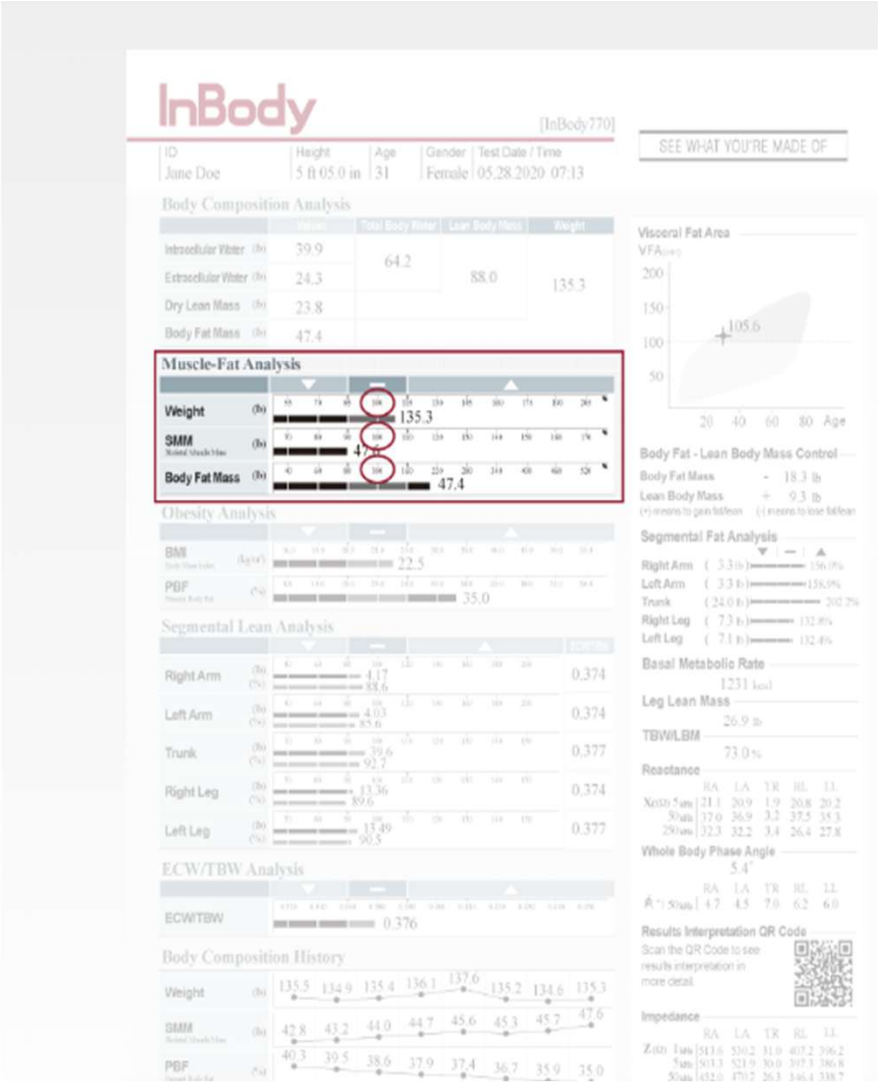
How to read the numbers at the top

You can compare your measurements to others of the same height and sex with the percentages above the bar graphs. While the healthy range varies by output, the 100% mark indicates the healthy average for people of your height and sex. So if your Weight bar is at 130%, this would mean that your weight is 30% above average.

Similarly, if your Weight bar is at 70%, this would mean that you have 30% less mass than is considered the healthy average for your height.

How to find your body type using the C-I-D Method

Refer to your Muscle-Fat Analysis and connect the endpoints of each bar (Weight to SMM to Body Fat Mass) to form a C, I, or D.



Muscle, Fat, & Obesity Risk

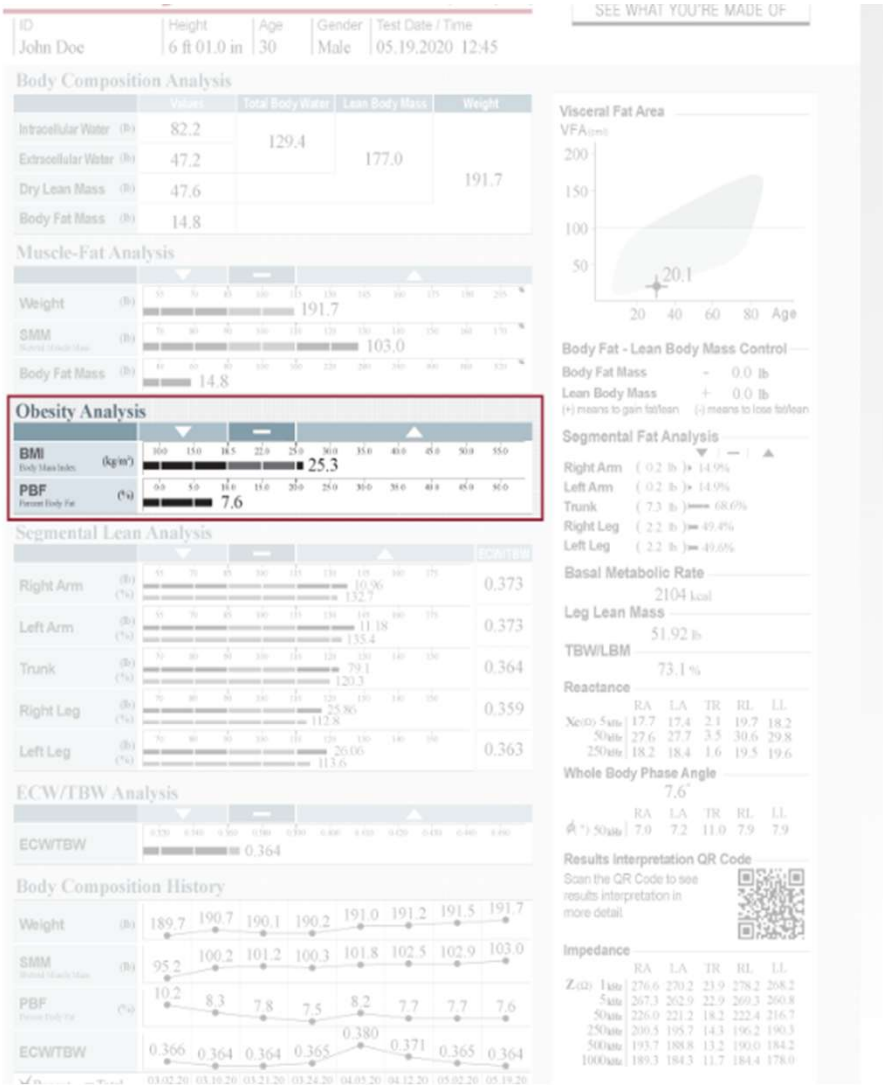
Obesity Analysis

This section displays your body fat percentage, or Percent Body Fat (PBF), and Body Mass Index (BMI), as well as the healthy ranges for these measurements.

Let's explore how Obesity Analysis can give you a better idea of your current obesity level and your risk of related health conditions.

Body fat percentage is a division of your body fat mass by your total weight and a much better indicator of your risk of obesity than BMI.

Let's take a closer look at why PBF is a superior metric for assessing your obesity and overall health risks.



Chapter 3: Segmental Lean Analysis—Your Magnifying Glass

Segmental Lean Analysis

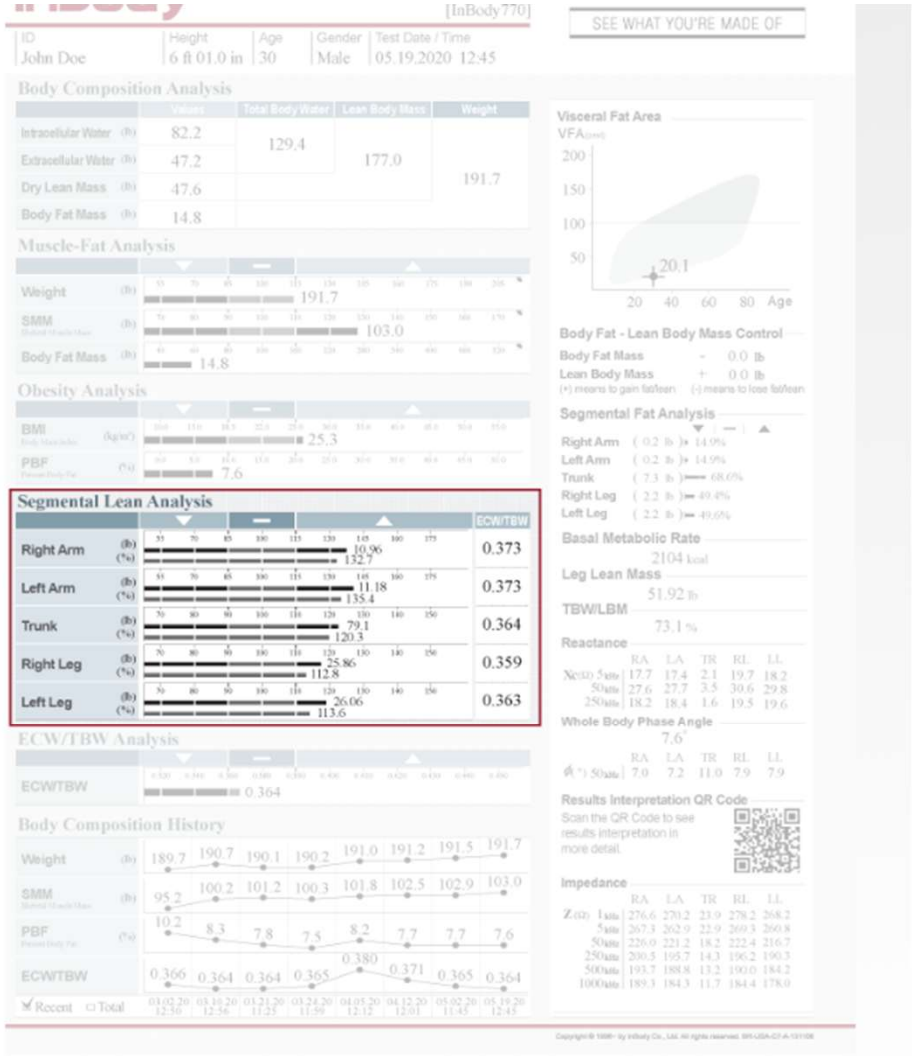
When used correctly, this is one of the most valuable sections of your InBody Result Sheet.

Segmental Lean Analysis displays your Lean Body Mass (Fat-Free Mass) in all body segments in pounds and its sufficiency to support your body weight as a percentage.

Interpreting your **Segmental Lean Analysis** can help you:

- Spot areas for improvement
- Assess if your Lean Body Mass is sufficiently developed in all body segments
- Identify if you have any muscle imbalances

The InBody divides your body into five body segments: your arms, legs, and trunk (torso), or the area between your neck, arms, and legs. The data for each body segment is displayed as two bars.



Body Composition Analysis & Body Water

ECW/TBW Analysis

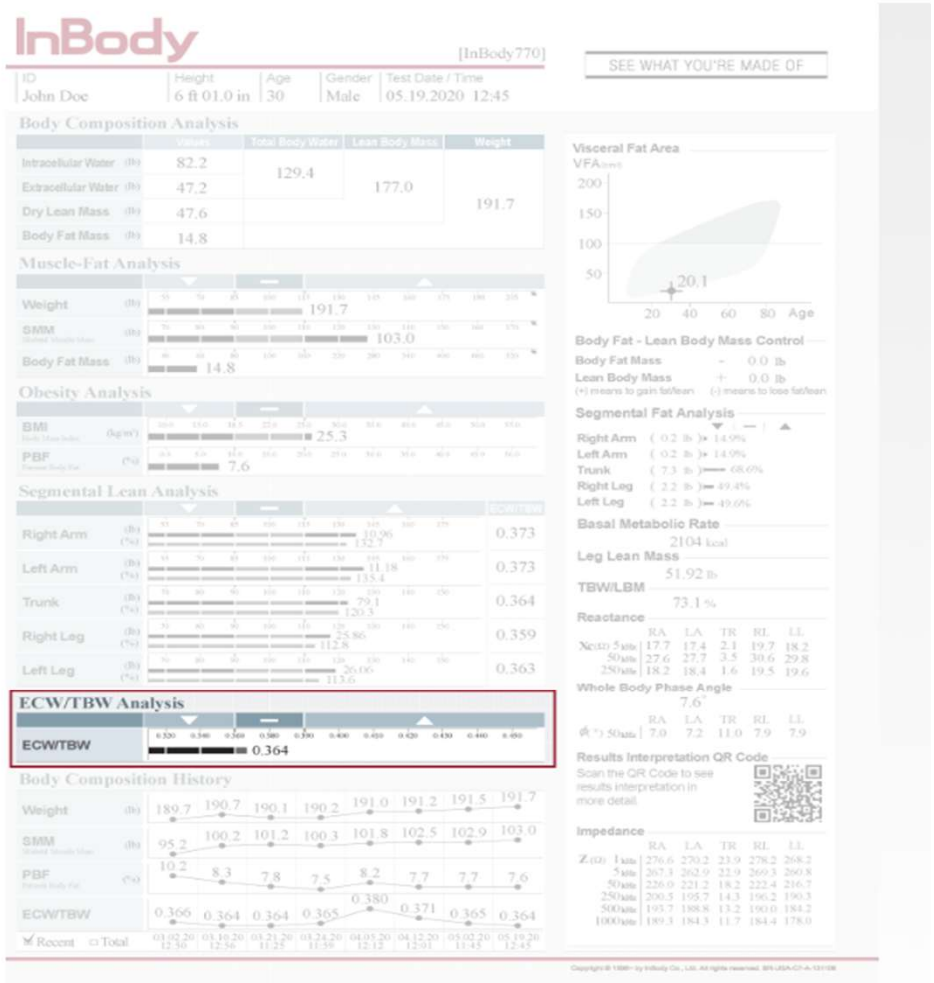
ECW/TBW is a measure of compartmental fluid distribution and shows, as a ratio, how much of your total body water is extracellular.

Taking multiple InBody Tests will establish your normal fluid status and help determine any imbalances. As a general guide, check to see that your ECW/TBW is below 0.390.

You'll see minor fluctuations in your ECW/TBW—that's normal. This output is primarily used to give context to another section—the Segmental Lean Analysis.

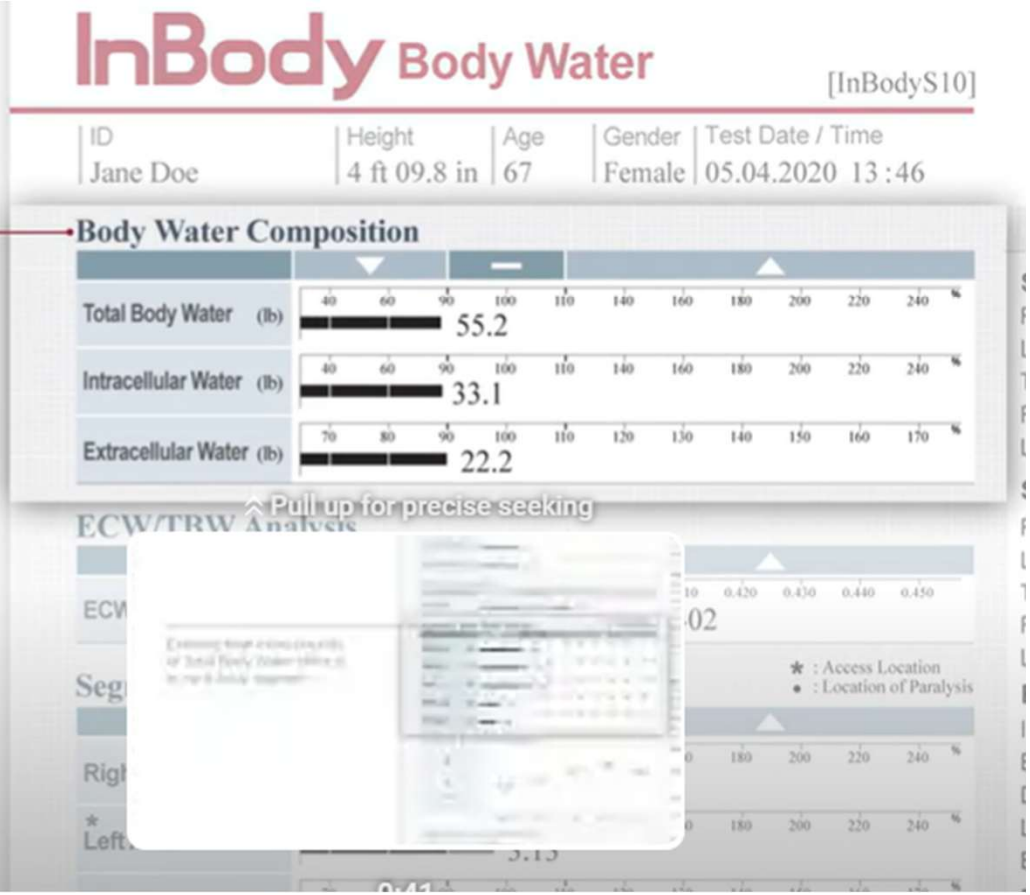
WHY THIS MATTERS

This graph lets you quickly understand if you have occasional inflammation or swelling in your body resulting from strenuous exercise or activity. You can also use this to give context to LBM. High LBM and a high ECW/TBW ratio usually indicate excess body water—not just muscle.

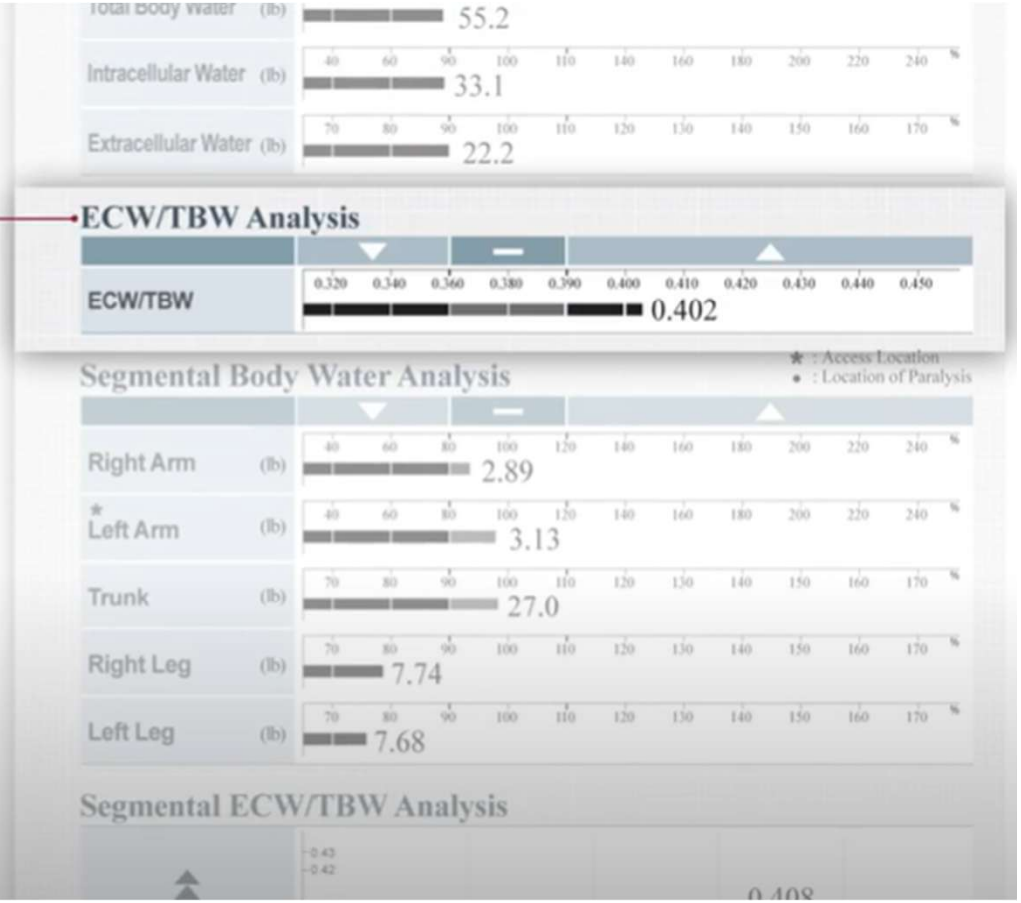


Body Water Analysis Sheet

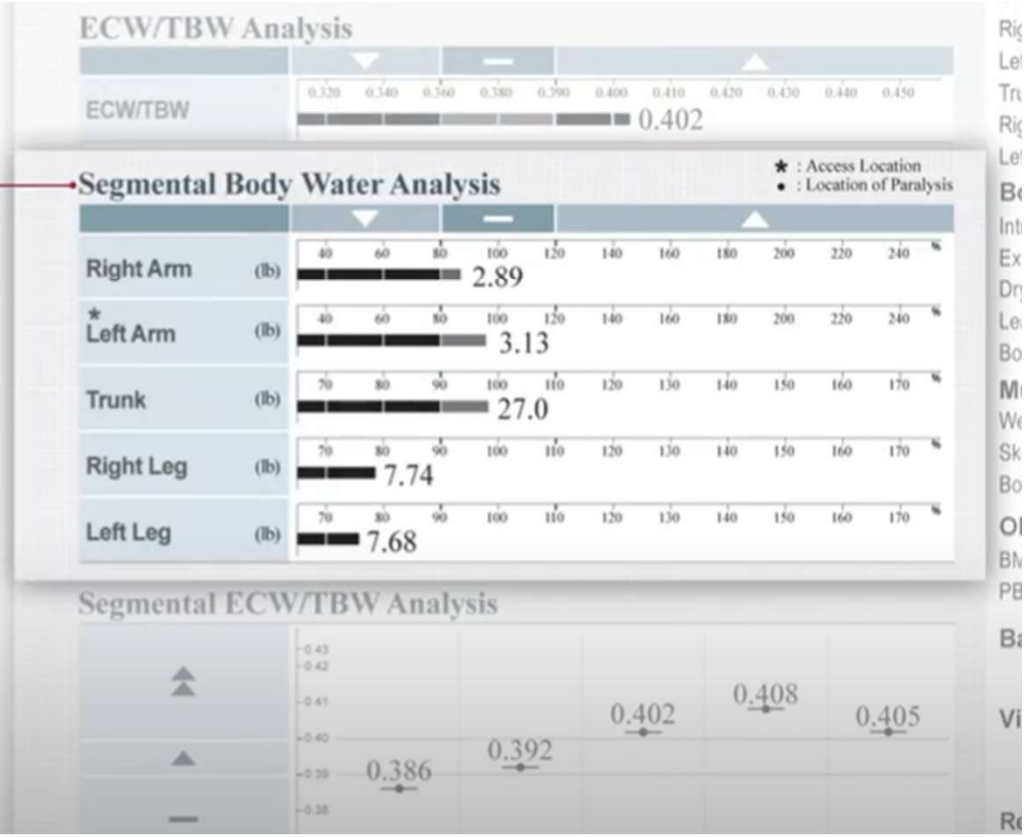
View Total Body Water in terms of weight to determine a healthy ICW/ECW balance



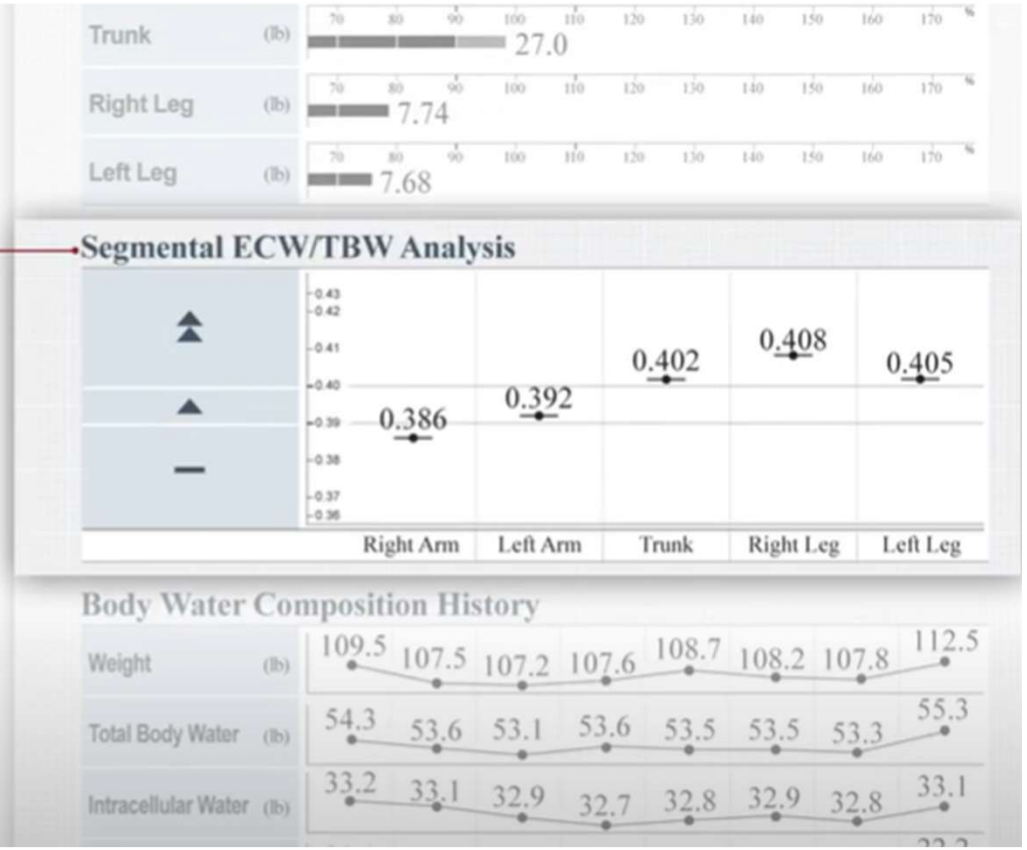
Spot unusual shifts in fluid distribution to help detect swelling, inflammation, & more



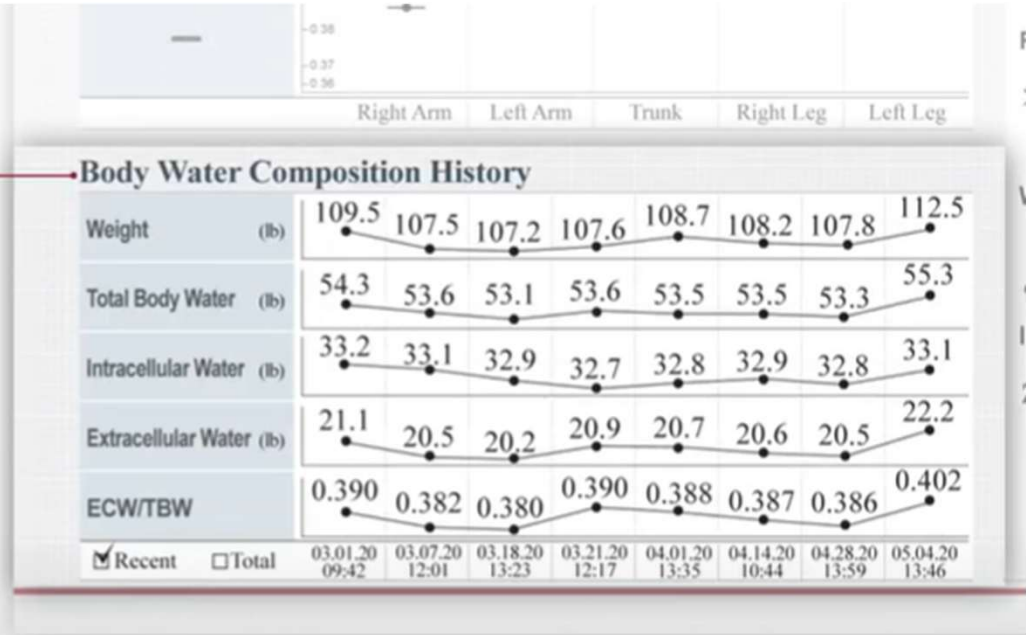
Examine how many pounds of Total Body Water there is in each body segment



See ECW/TBW in all five segments to help monitor localized swelling and assist with dialysis treatment



Track changes in body water over time to precisely monitor fluid balance and improve care



InBody

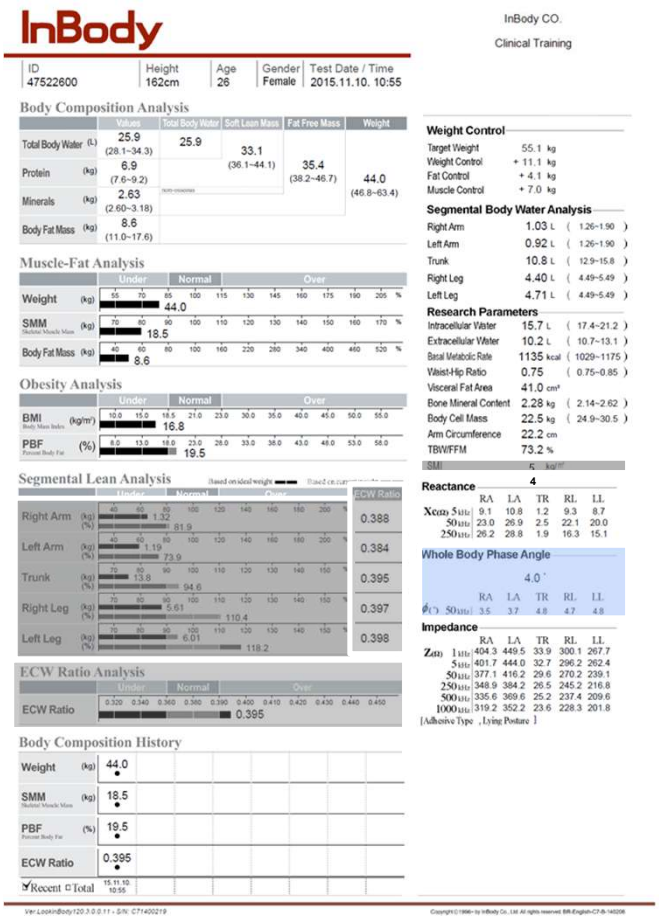




InBody S10

& Cardiac Rehabilitation





- Application in Cardiac Rehabilitation -

Edema Evaluation

- ECW Ratio: Monitoring Water balance
- Segmental ECW Ratio: Ascites Evaluation

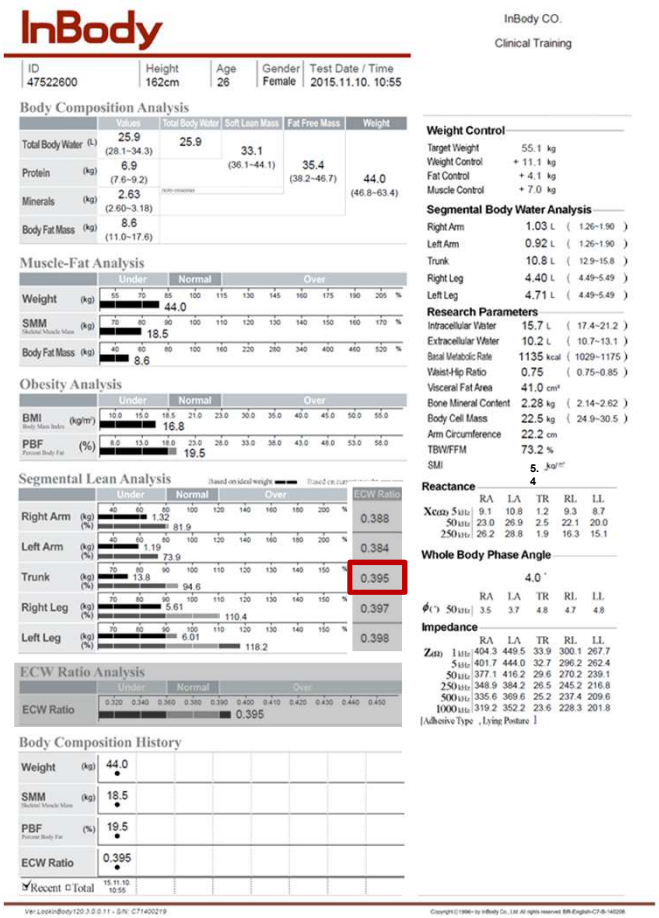
Sarcopenia Assessment & Rehab Monitoring

- SMI: Sarcopenia Assessment
- Segmental Lean: Monitoring Rehab Outcomes

Mortality & Nutritional Assessment

- Phase Angle: Severity & Nutrition Assessment





InBody

InBody CO.
Clinical Training

ID	Height	Age	Gender	Test Date / Time
47522600	162cm	26	Female	2015.11.10. 10:55

Body Composition Analysis

	Values	Total Body Water	Soft Lean Mass	Fat Free Mass	Weight
Total Body Water (L)	25.9 (28.1~34.3)	25.9			
Protein (kg)	6.9 (7.6~9.2)		33.1 (36.1~44.1)	35.4 (38.2~46.7)	44.0 (46.8~63.4)
Minerals (kg)	2.63 (2.60~3.18)				
Body Fat Mass (kg)	8.6 (11.0~17.6)				

Muscle-Fat Analysis

	Under	Normal	Over
Weight (kg)		44.0	
SMM (kg)		18.5	
Body Fat Mass (kg)		8.6	

Obesity Analysis

	Under	Normal	Over
BMI (kg/m²)		16.8	
PBF (%)		19.5	

Segmental Lean Analysis

	Under	Normal	Over	ECW Ratio
Right Arm (kg) (%)		1.32 (81.9)		0.388
Left Arm (kg) (%)		1.19 (73.9)		0.384
Trunk (kg) (%)		13.8 (94.6)		0.395
Right Leg (kg) (%)		5.61 (110.4)		0.397
Left Leg (kg) (%)		6.01 (118.2)		0.398

ECW Ratio Analysis

	Under	Normal	Over
ECW Ratio		0.395	

Body Composition History

Weight (kg)	44.0	
SMM (kg)	18.5	
PBF (%)	19.5	
ECW Ratio	0.395	

Recent Total 15.11.10 10:55

Weight Control

Target Weight	55.1 kg	
Weight Control	+ 11.1 kg	
Fat Control	+ 4.1 kg	
Muscle Control	+ 7.0 kg	

Segmental Body Water Analysis

Right Arm	1.03 L	(1.26~1.90)
Left Arm	0.92 L	(1.26~1.90)
Trunk	10.8 L	(12.9~15.8)
Right Leg	4.40 L	(4.49~5.49)
Left Leg	4.71 L	(4.49~5.49)

Research Parameters

Intracellular Water	15.7 L	(17.4~21.2)
Extracellular Water	10.2 L	(10.7~13.1)
Basal Metabolic Rate	1135 kcal	(1029~1175)
Waist-Hip Ratio	0.75	(0.75~0.85)
Visceral Fat Area	41.0 cm²	
Bone Mineral Content	2.28 kg	(2.14~2.62)
Body Cell Mass	22.5 kg	(24.9~30.5)
Arm Circumference	22.2 cm	
TBW/FFM	73.2 %	
SMI	5.4 kg/m²	

Reactance

	RA	LA	TR	RL	LL
Xc(Ω) 5 kHz	9.1	10.8	1.2	9.3	8.7
50 kHz	23.0	26.9	2.5	22.1	20.0
250 kHz	26.2	28.8	1.9	16.3	15.1

Whole Body Phase Angle

4.0°

	RA	LA	TR	RL	LL
φ(°) 50 kHz	3.5	3.7	4.8	4.7	4.8

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1 kHz	404.3	449.5	33.9	300.1	267.7
5 kHz	401.7	444.0	32.7	296.2	262.4
50 kHz	377.1	416.2	29.6	270.2	239.1
250 kHz	348.9	384.2	26.5	245.2	216.8
500 kHz	335.6	369.6	25.2	237.4	209.6
1000 kHz	319.2	352.2	23.6	228.3	201.8

[Adhesive Type , Lying Posture]

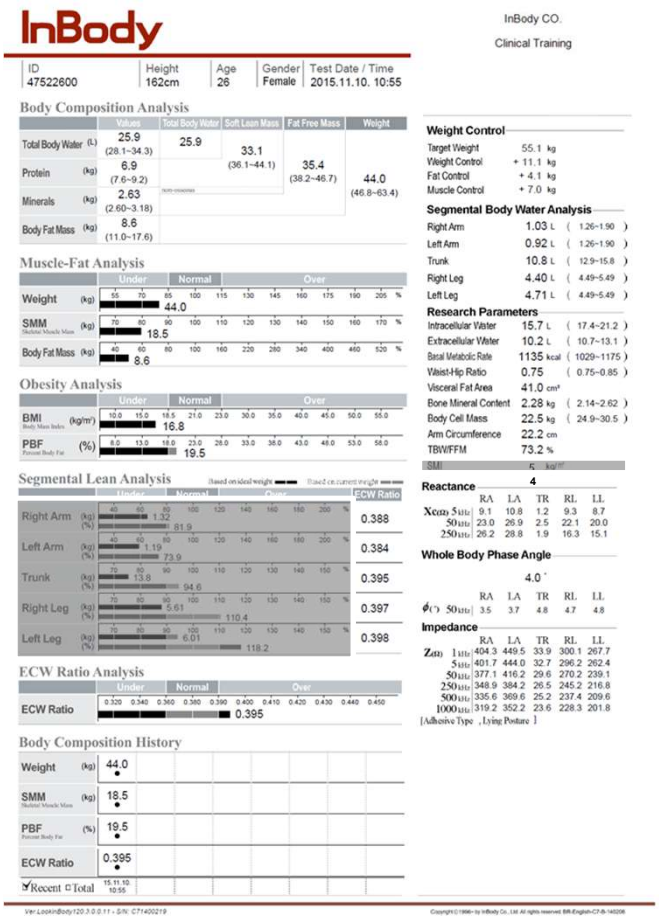
- Application in Cardiac Rehabilitation -

Edema Evaluation
- ECW Ratio: Monitoring Water balance - Segmental ECW Ratio: Ascites Evaluation

[ECW Ratio Evaluation]

ECW Ratio (ECW/TBW)
Normal: 0.360-0.390
Slight Edema: 0.390-0.400
Severe Edema: Over 0.400





InBody

InBody CO.
Clinical Training

ID	Height	Age	Gender	Test Date / Time
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Total Body Water (L)	25.9 (28.1~34.3)	25.9	33.1 (36.1~44.1)	35.4 (38.2~46.7)	44.0 (46.8~63.4)
Protein (kg)	6.9 (7.6~9.2)				
Minerals (kg)	2.63 (2.60~3.18)				
Body Fat Mass (kg)	8.6 (11.0~17.6)				

Muscle-Fat Analysis

	Under	Normal	Over
Weight (kg)		44.0	
SMM (kg)		18.5	
Body Fat Mass (kg)		8.6	

Obesity Analysis

	Under	Normal	Over
BMI (kg/m²)		16.8	
PBF (%)		19.5	

Segmental Lean Analysis

	Under	Normal	Over	ECW Ratio
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Right Leg (kg) (%)		5.61 (110.4)		0.397
Left Leg (kg) (%)		6.01 (118.2)		0.398

ECW Ratio Analysis

	Under	Normal	Over
ECW Ratio		0.395	

Body Composition History

Weight (kg)	44.0
SMM (kg)	18.5
PBF (%)	19.5
ECW Ratio	0.395

Recent / Total 15.11.10 10:55

Weight Control

Target Weight	55.1 kg
Weight Control	+ 11.1 kg
Fat Control	+ 4.1 kg
Muscle Control	+ 7.0 kg

Segmental Body Water Analysis

Right Arm	1.03 L	(1.26~1.90)
Left Arm	0.92 L	(1.26~1.90)
Trunk	10.8 L	(12.9~15.8)
Right Leg	4.40 L	(4.49~5.49)
Left Leg	4.71 L	(4.49~5.49)

Research Parameters

Intracellular Water	15.7 L	(17.4~21.2)
Extracellular Water	10.2 L	(10.7~13.1)
Basal Metabolic Rate	1135 kcal	(1029~1175)
Waist-Hip Ratio	0.75	(0.75~0.85)
Visceral Fat Area	41.0 cm²	
Bone Mineral Content	2.28 kg	(2.14~2.62)
Body Cell Mass	22.5 kg	(24.9~30.5)
Arm Circumference	22.2 cm	
TBW/FFM	73.2 %	
SLI	5 kg/m²	

Reactance (4)

	RA	LA	TR	RL	LL
Xc(Ω) 5 kHz	9.1	10.8	1.2	9.3	8.7
50 kHz	23.0	26.9	2.5	22.1	20.0
250 kHz	26.2	28.8	1.9	16.3	15.1

Whole Body Phase Angle

4.0°

	RA	LA	TR	RL	LL
φ(°) 50 kHz	3.5	3.7	4.8	4.7	4.8

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1 kHz	404.3	449.5	33.9	300.1	267.7
5 kHz	401.7	444.0	32.7	296.2	262.4
50 kHz	377.1	416.2	29.6	270.2	239.1
250 kHz	348.9	384.2	26.5	245.2	216.8
500 kHz	335.6	369.6	25.2	237.4	209.6
1000 kHz	319.2	352.2	23.6	228.3	201.8

[Adhesive Type , Lying Posture]

Ver:LookInBody120.3.0.0.11 - S/N: C7140021P

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- Application in Cardiac Rehabilitation -

Sarcopenia Assessment & Rehab Monitoring

- SMI: Sarcopenia Assessment
- Segmental Lean: Monitoring Rehab Outcomes

[Sarcopenia Assessment]

SMI

= appendicular skeletal muscle mass/height(m)^2
= (RA+LA+RL+LL Lean Mass)/height(m)^2

EWGSOP Cut-off

Cut-off points by gender	Reference group defined
Skeletal muscle mass index (SMI) (Appendicular skeletal muscle mass/height^2)	Based on 2 SD below mean of young adults (Rosetta Study)
Men: 7.26 kg/m^2	
Women: 5.5 kg/m^2	
SMI (ASM/height^2)	Based on sex-specific lowest 20% of study group (n = 2,976)
Men: 7.25 kg/m^2	
Women: 5.67 kg/m^2	
SMI (ASM/height^2)	Based on sex-specific lowest 20% (Health ABC Study)
Men: 7.23 kg/m^2	
Women: 5.67 kg/m^2	
Residuals of linear regression on appendicular lean mass adjusted for fat mass as well as height	Based on sex-specific lowest 20% (Health ABC Study)
Men: −2.29	
Women: −1.73	



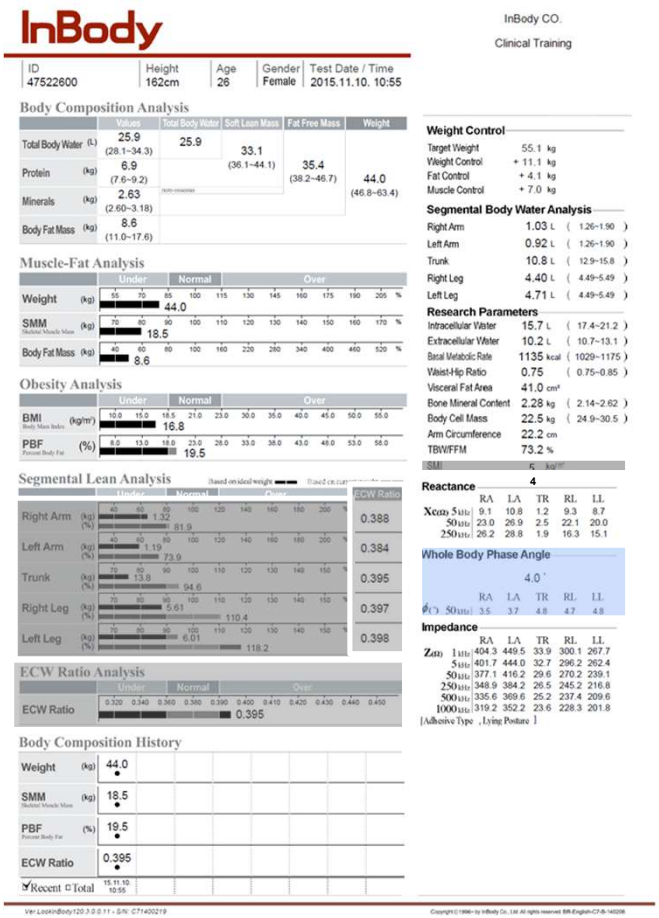
EFFECTIVENESS OF CARDIAC REHABILITATION FOR PREVENTION AND TREATMENT OF SARCOPENIA IN PATIENTS WITH CARDIOVASCULAR DISEASE - A RETROSPECTIVE CROSS-SECTIONAL ANALYSIS

H. HARADA[1], H. KAI[1], H. NIIYAMA[1], Y. NISHIYAMA[1], A. KATOH[1], N. YOSHIDA[2], Y. FUKUMOTO[3], H. IKEDA[4]

1. Department of Cardiology, Kurume University Medical Center, Kurume, Japan; 2. Institute of Health and Sports Sciences, Kurume University, Kurume, Japan; 3. Department of Cardio-vascular Medicine, Kurume University School of Medicine, Kurume, Japan; 4. Department of Physical Therapy, Faculty of Fukuoka Medical Technology, Teikyo University, Omuta, Japan. Corresponding author: Hisao Ikeda, MD, PhD, Department of Physical Therapy, Faculty of Fukuoka Medical Technology, Teikyo University, 6-22 Misaki-machi, Omuta, Fukuoka 836-8505, Japan, E-mail: ikeda_hisao@fmt.teikyo-u.ac.jp, Tel: +81-944-57-8333, Fax: +81-944-55-7703

Abstract: *Objective:* Sarcopenia is a syndrome characterized by progressive and generalized loss of skeletal muscle mass and strength, with the risk of frailty and poor quality of life. This study aimed to clarify the clinical characteristics of sarcopenia and to investigate the effects of comprehensive cardiac rehabilitation (CCR), including nutrition, physical exercise and medication, in patients with cardiovascular disease (CVD). *Methods:* We retrospectively studied 322 inpatients with CVD (age 72±12 years). Muscle mass, muscle strength and physical performance were assessed before and after exercise training in patients with and without sarcopenia, which was defined as either a gait speed of <0.8 m/s or reduced handgrip strength (<26 kg in males and <18 kg in females), together with lower skeletal muscle index (SMI) (<7.0 kg/m^2 in males and <5.7 kg/m^2 in females). The actual daily total calorie and nutrient intake was also calculated. *Results:* Sarcopenia was identified in 28% of patients with CVD, these patients having a higher prevalence of symptomatic chronic heart failure and chronic kidney disease. SMI was significantly associated with protein intake and statin treatment. The ratio of peak VO_2 and SMI was significantly higher in the statin treatment group. Handgrip strength, gait speed, leg weight bearing index, and nutritional intake improved after exercise training in patients both with and without sarcopenia. *Conclusions:* The present findings suggest that CCR is a promising strategy for prevention and treatment of sarcopenia in patients with CVD.





InBody

InBody CO.
Clinical Training

ID	Height	Age	Gender	Test Date / Time
47522600	162cm	26	Female	2015.11.10. 10:55

Body Composition Analysis

	Values	Total Body Water	Soft Lean Mass	Fat Free Mass	Weight
Total Body Water (L)	25.9 (28.1~34.3)	25.9			
Protein (kg)	6.9 (7.6~9.2)		33.1 (36.1~44.1)	35.4 (38.2~46.7)	44.0 (46.8~63.4)
Minerals (kg)	2.63 (2.60~3.18)				
Body Fat Mass (kg)	8.6 (11.0~17.6)				

Muscle-Fat Analysis

	Under	Normal	Over
Weight (kg)		44.0	
SMM (kg)		18.5	
Body Fat Mass (kg)		8.6	

Obesity Analysis

	Under	Normal	Over
BMI (kg/m²)		16.8	
PBF (%)		19.5	

Segmental Lean Analysis

	Under	Normal	Over	ECW Ratio
Right Arm (kg) (%)		1.32 (81.9)		0.388
Left Arm (kg) (%)		1.19 (73.9)		0.384
Trunk (kg) (%)		13.8 (94.6)		0.395
Right Leg (kg) (%)		5.61 (110.4)		0.397
Left Leg (kg) (%)		6.01 (118.2)		0.398

ECW Ratio Analysis

	Under	Normal	Over
ECW Ratio		0.395	

Body Composition History

Weight (kg)	44.0	
SMM (kg)	18.5	
PBF (%)	19.5	
ECW Ratio	0.395	

Recent Total 15.11.10 10:55

Weight Control

Target Weight	55.1 kg	
Weight Control	+ 11.1 kg	
Fat Control	+ 4.1 kg	
Muscle Control	+ 7.0 kg	

Segmental Body Water Analysis

Right Arm	1.03 L	(1.26~1.90)
Left Arm	0.92 L	(1.26~1.90)
Trunk	10.8 L	(12.9~15.8)
Right Leg	4.40 L	(4.49~5.49)
Left Leg	4.71 L	(4.49~5.49)

Research Parameters

Intracellular Water	15.7 L	(17.4~21.2)
Extracellular Water	10.2 L	(10.7~13.1)
Basal Metabolic Rate	1135 kcal	(1029~1175)
Waist-Hip Ratio	0.75	(0.75~0.85)
Visceral Fat Area	41.0 cm²	
Bone Mineral Content	2.28 kg	(2.14~2.62)
Body Cell Mass	22.5 kg	(24.9~30.5)
Arm Circumference	22.2 cm	
TBW/FFM	73.2 %	
SLI	5 kg/m²	

Reactance

	RA	LA	TR	RL	LL
Xc(Ω) 5 kHz	9.1	10.8	1.2	9.3	8.7
50 kHz	23.0	26.9	2.5	22.1	20.0
250 kHz	26.2	28.8	1.9	16.3	15.1

Whole Body Phase Angle 4.0°

	RA	LA	TR	RL	LL
Φ(°) 50 kHz	3.5	3.7	4.8	4.7	4.8

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1 kHz	404.3	449.5	33.9	300.1	267.7
5 kHz	401.7	444.0	32.7	296.2	262.4
50 kHz	377.1	416.2	29.6	270.2	239.1
250 kHz	348.9	384.2	26.5	245.2	216.8
500 kHz	335.6	369.6	25.2	237.4	209.6
1000 kHz	319.2	352.2	23.6	228.3	201.8

[Adhesive Type , Lying Posture]

Ver:LookInBody120.3.0.0.11 - S/N: C7140021P

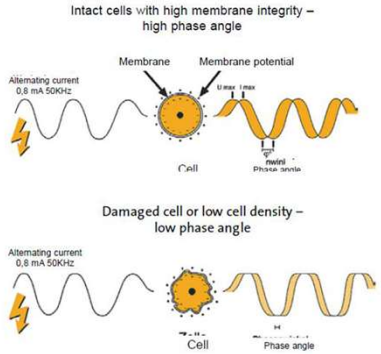
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- Application in Cardiac Rehabilitation -

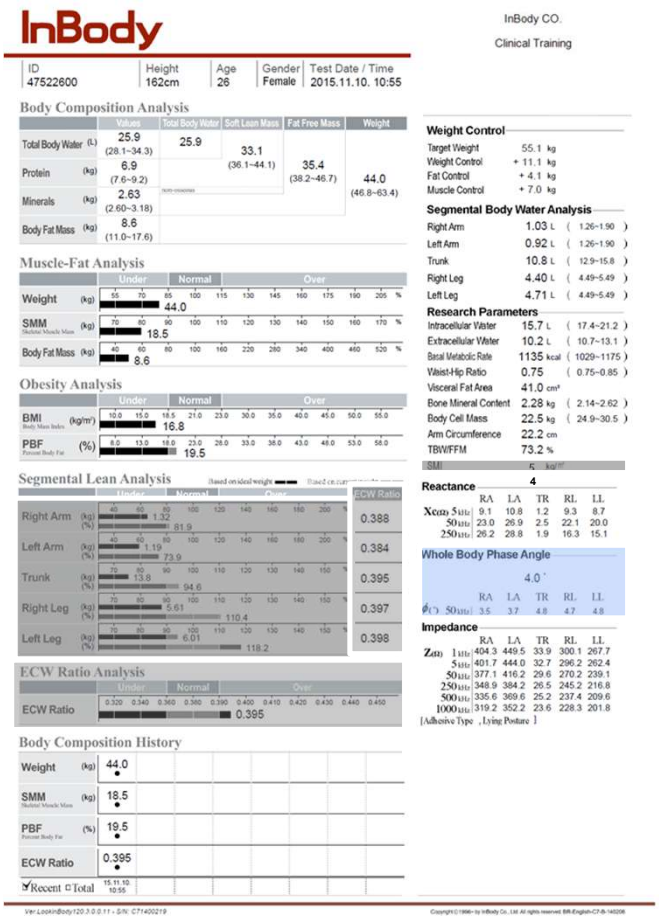
Mortality & Nutritional Assessment

- Phase Angle: Severity & Nutrition Assessment

Intensity & Healthiness of cellular membrane makes Phase Angle higher, whereas malnutrition & aging can cause Phase Angle decreased.







InBody

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Clinical Training

ID	Height	Age	Gender	Test Date / Time
47522600	162cm	26	Female	2015.11.10. 10:55

Body Composition Analysis

	Values	Total Body Water	Soft Lean Mass	Fat Free Mass	Weight
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Research Parameters

Intracellular Water	15.7 L	(17.4~21.2)
Extracellular Water	10.2 L	(10.7~13.1)
Basal Metabolic Rate	1135 kcal	(1029~1175)
Waist-Hip Ratio	0.75	(0.75~0.85)
Visceral Fat Area	41.0 cm²	
Bone Mineral Content	2.28 kg	(2.14~2.62)
Body Cell Mass	22.5 kg	(24.9~30.5)
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TBW/FFM	73.2 %	
SLI	5 kg/m²	

Reactance

	RA	LA	TR	RL	LL
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50 kHz	23.0	26.9	2.5	22.1	20.0
250 kHz	26.2	28.8	1.9	16.3	15.1

Whole Body Phase Angle

4.0°

	RA	LA	TR	RL	LL
φ(°) 50 kHz	3.5	3.7	4.8	4.7	4.8

Impedance

	RA	LA	TR	RL	LL
Z(Ω) 1 kHz	404.3	449.5	33.9	300.1	267.7
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[Adhesive Type , Lying Posture]

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- Application in Cardiac Rehabilitation -

Mortality & Nutritional Assessment

- Phase Angle: Severity & Nutrition Assessment

[Phase Angle]

Normal: Over 4
Consideration of age, Medical History: 2 ~ 4
Severe Status: Under 2

InBody [Phase Angle & Nutrition]



[IB S10]

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ORIGINAL RESEARCH

Malnutrition Assessment in Hemodialysis Patients: Role of Bioelectrical Impedance Analysis Phase Angle

Laurynas Rimsevicius, MD, PhD,† Alvita Gincaite, MD,* Vaidas Vicka, MD,* Diana Sukackiene, MD,† Jelena Pavinic, MD,† and Marius Miglinas, MD, PhD*†*

Objective: To determine the most potent bioelectrical impedance analysis (BIA) marker of malnutrition and to adjust its application to hemodialysis (HD) patients.

Design: An observational study.

Subjects: A total of 99 patients on maintenance HD were enrolled in the study.

Intervention: The nutritional state of the patients was examined before and after the HD procedure using Subjective Global Assessment Scale (SGA), serum albumin, body mass index and BIA-derived fat-free mass index, reactance, resistance, and phase angle (PA). Malnutrition defined by the SGA questionnaire was used to detect the most potent marker of malnutrition. This marker was further analyzed and corrected for the excess fluid, age, and gender producing the nutritional state-specific cutoffs.

Results: The SGA rates of nutritional state were as follows: 57.6% (57) well nourished, 28.3% (28) moderately malnourished, and 14.1% (14) severely malnourished. Multivariate forward logistic regression analysis of the nutrition markers revealed PA as the most potent malnutrition predictor (odds ratio = 3.69; 95% confidence interval [CI]: 1.59-8.62; $P = .002$). PA was adjusted for the excess fluid (5.00 ± 0.97 vs 4.87 ± 1.08 $P < .001$). Patients were assigned into groups with adjusted PA values below the 5th through the 50th percentile of the mean PA reference value. The moderately malnourished patients were most accurately identified by the percentile group of <25th (area under the curve = 0.70; 95% CI: 0.60-0.81; $P = .001$), and the severely malnourished patients were most accurately identified by the percentile group of <15th (area under the curve = 0.74; 95% CI: 0.62-0.85; $P = .005$).

Conclusion: Malnutrition is present almost in a half of the HD patients. BIA-provided PA is the most potent predictor of malnutrition. PA can be adjusted for the excess fluid after HD, age, and gender and used accordingly.

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Introduction

MALNUTRITION IS A prevalent condition in patients experiencing various chronic diseases.[1] From 20% to 70% of the maintenance hemodialysis (HD) patients are found to be malnourished depending on the method used to evaluate the nutritional status.[2-4] This complex state is caused by factors comprising of uremia, persistent inflammation, physical disability, dietary restrictions, gastrointestinal tract disturbances, metabolic acidosis, or HD by itself.[5] Malnutrition is independently responsible for a decreased quality of life and a greater risk of mortality.[4] Therefore, early recognition and diagnosis of malnutrition are some of the key points in HD patients' management.

Bioelectrical impedance analysis (BIA) is widely recognized as a method for evaluating body composition, which contributes information about the nutritional status of the patient. BIA provides various nutritional markers, such as fat-free mass index, reactance, resistance, and phase angle (PA).[6] These markers are commonly studied, and their application is well established in cardiology,[7] cardiac surgery,[8] oncology,[9] pulmonology, and geriatrics.[10,11] BIA-derived nutritional markers are used to identify the malnourished patients and, therefore, to evaluate the risk of the deleterious clinical outcomes and provide a basis for the nutritional therapy.

In HD patients, BIA is mostly used for the evaluation of fluid balance. Application of BIA to diagnose malnutrition is challenging, and hence, criticized a lot.[12] The alterations in fluid and electrolyte balance distort the electric measurements and fault the results of BIA.[13] Overhydration,[14] age, and gender tend to decrease the bioelectrical impedance and thus cause false results leading to inaccurate diagnosis of malnutrition.[15] However, various BIA indices of

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Financial Disclosure: The authors declare that they have no relevant financial interests.

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1051-2276/$36.00
http://dx.doi.org/10.1053/j.jrn.2016.03.004

Table 2. Multivariate Regression Analysis of the Malnutrition Predictors

	Odds Ratio			Odds Ratio		
	Estimate	95% CI		Estimate	95% CI	
Factor	Univariate		*P* value	Multivariate		*P* Value
PA	3.70	1.79-7.69	<.001	3.69	1.59-8.62	.002
FFMI	1.28	0.96-1.69	.086	n.i.		
Reactance	1.16	1.03-1.32	.014	n.s.		
Resistance	1.00	0.98-1.01	.637	n.i.		
BMI	1.09	0.96-1.22	.195	n.i.		
Serum albumin	1.22	1.04-1.42	.013	n.s.		

BMI, body mass index; CI, confidence interval; FFMI, fat-free mass index; n.i., not included; n.s., not significant with *P* value >.05; PA, phase angle.

Conclusion

Malnutrition is present in almost a half of the HD patients. We found that BIA-provided PA is an accurate and appropriate method of nutritional state evaluation. Furthermore, despite excess fluid, age and gender being important limitations in the application of BIA, PA can be adjusted for these factors and used accordingly. The malnutrition cutoff percentiles of the PA we report in our study must be further evaluated in upcoming research.

InBody [Phase Angle & Outcomes after cardiac surgery] [IB S10]



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Original article

Malnutrition assessed by phase angle determines outcomes in low-risk cardiac surgery patients

Donata Ringaitiene [a, f, *], Dalia Gineityte [b], Vaidas Vicka [b], Tadas Zvirblis [c], Ieva Norkiene [a, f], Jurate Sipylaite [a, f], Algimantas Irnius [d, e], Juozas Ivaskevicius [a]

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Cardiac surgery
Phase angle
Bioelectrical impedance analysis
Postoperative outcomes

SUMMARY

Background & aims: Phase angle (PA), which is obtained from bioelectrical impedance analysis (BIA), is a non-invasive method for measuring altered electrical properties of biological tissues. It has been recognised as an objective prognostic marker of disease severity and frailty. The aim of this study is to determine whether PA is a marker of malnutrition and postoperative morbidity in low operative risk patients undergoing cardiac surgery.

Methods: A prospective study was conducted in a tertiary hospital. The nutritional state of the cardiac surgery patients was evaluated using BIA the day before the scheduled surgery. After applying selection criteria, 342 low operative risk patients were selected and classified into two groups in accordance with the PA value: a low PA group and a normal PA group. The correlation between low PA and low fat-free mass index (FFMI), a marker of malnutrition, was assessed. Associations between low PA and adverse postoperative outcomes, defined by the Society of Thoracic Surgeons postoperative risk evaluation model, were analysed. The impact of low PA on length of stay in an ICU and hospital was evaluated.

Results: Low PA was detected in 61 (17.8%) patients in the selected group, which consisted of low operative risk patients with a median Euroscore II value of 1.46 (IQR: 0.97–2.03) and was associated with FFMI with Pearson's R of 0.515 ($p < 0.001$). Low PA was associated with higher rates (13 [21.3%] vs. 30 [10.7%] $p = 0.023$) and risk of postoperative morbidity in univariate regression analysis (OR = 2.27, CI 95%: 1.10–4.66, $p = 0.026$). Furthermore, low PA persisted as an independent factor in multivariate regression analysis (OR = 2.50, CI 95%: 1.18–5.29, $p = 0.016$) adjusted for preoperative risk factors of postoperative morbidity. Evaluation of hospitalisation length revealed a tendency of a prolonged hospitalisation (>14 days) rate (31 [50.8%] vs. 105 [37.8%], $p = 0.063$) in the group with low PA.

Conclusion: A low preoperative PA is an indicator of malnutrition and determines adverse outcomes after cardiac surgery. Further research is needed to evaluate clinical applications of the PA, such as a more accurate identification of malnourished cardiac surgery patients.

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Abbreviations: PA, phase angle; BIA, bioelectrical impedance analysis; Low PA, low phase angle; FFMI, fat-free mass index; STS, Society of Thoracic Surgeons; ICU, intensive care unit; Euroscore II, European system for cardiac operative risk evaluation II; IQR, interquartile range; OR, odds ratio; CI, confidence interval; BMI, body mass index; CABG, coronary artery bypass grafting; LVEF, left ventricle ejection fraction; IABP, intra-aortic aortic balloon pump; ESPEN, European Society of Clinical Nutrition and Metabolism; PLOS, perioperative length of stay; CPB, cardiopulmonary bypass; CDC, Centre for Disease Control and Prevention; ROC, receiver operating characteristic; AUC, area under the curve; NYHA, New York Heart Association Functional classification.

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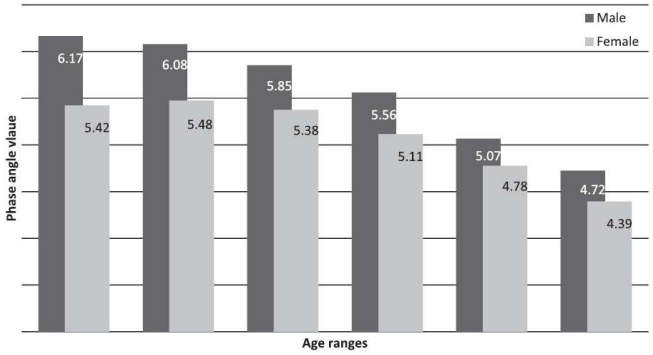


Fig. 1. Phase angle population reference values of the 15th percentile for men and women according to Bosy-Westphal et al. [20].

3.3. Low PA in relation to clinical outcomes

The rate of postoperative morbidity was 43 (12.6%) in the study cohort. The prevalence of postoperative morbidity was two times higher in the low PA group than in the normal PA group (13 [21.3%] vs. 30 [10.7%], $p = 0.023$, respectively). Low PA, CPB time, mitral valve repair, and combined type of surgery were found to be predictors of postoperative STS morbidity in the univariate analysis. Mitral valve repair, combined type of surgery, and low PA persisted as independent predictive factors for STS morbidity in the multivariate regression model (Table 2).

Evaluation of hospitalisation length showed a tendency of prolonged postoperative hospitalisation (>14 days) rate (31 [50.8%] vs. 105 [37.8%], $p = 0.063$) in the group with low PA. Further analysis of hospitalisation lengths revealed statistically significant differences in total time spent in the hospital postoperatively (median [IQR]: 14 [11–15] vs. 12 [11–14], $p = 0.036$) but no difference in the length of stay in the ICU (median [IQR]: 2 [2–4] vs. 2 [1–3], $p = 0.106$).



ORIGINAL RESEARCH

Phase Angle as a Biomarker for Frailty and Postoperative Mortality: The BICS Study

Louis Mullie, MD; Alexandrine Obrand, BSc; Melissa Bendayan, BSc; Amanda Trnkus, MSc; Marie-Claude Ouimet, BSc; Emmanuel Moss, MD; Annabel Chen-Tournoux, MD; Lawrence G. Rudski, MD, MSc; Jonathan Afilalo, MD, MSc

Background—Phase angle (PA) is a bioimpedance measurement that is determined lean body mass and hydration s with low PA values are more likely to be frail, sarcopenic, or malnourished. Previous work has shown that low PA pr outcomes after cardiac surgery, but the effect of PA on survival has not previously been assessed in this setting.

Methods and Results—The BICS (Bioimpedance in Cardiac Surgery) study recruited 277 patients undergoing surgery at 2 university-affiliated hospitals in Montreal, QC, Canada. Bioimpedance measurements as well as frailty assessments were performed preoperatively. The primary outcome was all-cause mortality. Secondary outcome mortality, postoperative morbidity, and hospital length of stay. There were 10 deaths at 1 month of follow-up and 12 months of follow-up. PA was associated with age, sex, body mass index, comorbidities, and frailty, as measure Physical Performance Battery and Fried scales. After adjusting for Society of Thoracic Surgeons–predicted mortality, associated with higher mortality at 1 month (adjusted odds ratio, 3.57 per 1° decrease in PA; 95% confidence interv and at 12 months (adjusted odds ratio, 3.03 per 1° decrease in PA; 95% confidence interval, 1.30–7.09), a higher morbidity (adjusted hazard ratio, 2.51 per 1° decrease in PA; 95% confidence interval, 1.32–4.75), and a longer hos stay (adjusted β, 4.8 days per 1° decrease in PA; 95% confidence interval, 1.3–8.2 days).

Conclusions—Low PA is associated with frailty and is predictive of mortality, morbidity, and length of stay after surgery. Further work is needed to determine the responsiveness of PA to interventions aimed at reversing frailty.
***Assoc.* 2018;7:e008721. DOI: 10.1161/JAHA.118.008721.)**

Key Words: biompedance • cardiac surgery • frailty • phase angle • sarcopenia

Frailty is a geriatric syndrome that has demonstrated prognostic significance in older adults undergoing cardiac surgery.[1] Although numerous clinical scales exist to evaluate frailty and predict outcomes, many scales are time-consuming to administer, motivating the search for objective

biomarkers that can serve as specific indica
Bioimpedance is a noninvasive method for bod
assessment that has been extensively validated
of body composition[2–6] and hydration status[7]
undergoing cardiac surgery. Phase angle (PA)
ment obtained via bioimpedance analysis thr
resistance and reactance of the human body
the application of an external current.[14] P
associated with sarcopenia[15–18] and frailty[16]
patients, and represents a promising candidat
vasive tool for frailty assessment in the cardia

PA is determined by lean body mass, total bod
and the ratio of extracellular water (ECW) to TBI
decrease in muscle mass or an increase in ECV
decrease PA. In cardiac patients, low PA is asso
muscle mass and handgrip strength.[21,22] In the s
failure, low PA is associated with increased mo
prolonged length of stay (LOS).[24] Thus, PA incorpo
mass and hydration status into a noninvasive ma
suited to the evaluation of cardiac patients. PA ca
<1 minute at the bedside, and contrary to other

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Table 4. Postoperative Outcomes According to PA Tertile

Outcome	Total	PA ≤4.5°	PA 4.6°–5.5°	PA ≥5.6°	*P* Value
STS predicted risk					
Mortality, %	1.9±1.9	3.0±2.6	1.8±1.3	1.0±0.7	<0.001
Morbidity or mortality, %	14.3±8.1	18.6±10.2	13.9±6.3	10.3±4.5	<0.001
Long length of stay, %	6.1±4.7	8.7±6.0	5.8±3.7	3.7±2.2	<0.001
Postoperative outcomes					
ICU length of stay, h	57±119	76±171	61±111	34±35	<0.001
Hospital length of stay, d	13±23	21±37	10±11	9±6	<0.001
Discharge to rehabilitation or nursing home	64 (23)	34 (38)	21 (20)	9 (11)	<0.001
Death at 1 mo	10 (4)	7 (8)	3 (3)	0 (0)	0.006
Death at 12 mo	16 (9)	11 (19)	5 (7)	0 (0)	<0.001
Readmission at 1 mo	38 (14)	15 (18)	12 (12)	11 (13)	0.43
Readmission at 12 mo	39 (26)	19 (43)	10 (17)	10 (20)	0.02
STS composite major morbidity	70 (25)	33 (37)	28 (27)	9 (11)	<0.001
Postoperative complications					
Stroke	4 (1)	1 (1)	1 (1)	2 (2)	0.49
Sepsis	6 (2)	4 (4)	1 (1)	1 (1)	0.13
Prolonged ventilation	34 (12)	15 (17)	13 (12)	6 (7)	0.06
Acute kidney injury	58 (21)	27 (30)	21 (20)	10 (12)	0.003
Dialysis	3 (1)	3 (3)	0 (0)	0 (0)	0.03
Myocardial infarction	4 (1)	2 (2)	2 (2)	0 (0)	0.22
Bleeding	59 (21)	26 (29)	24 (23)	9 (11)	0.003
Vascular complication	12 (4)	7 (8)	3 (3)	2 (2)	0.08
Atrial fibrillation	90 (32)	30 (34)	36 (34)	24 (29)	0.51
Cardiac arrest	10 (4)	5 (6)	4 (4)	1 (1)	0.12
Delirium	56 (23)	25 (32)	20 (21)	11 (14)	0.007

Data are given as mean±SD or number (percentage). ICU indicates intensive care unit; PA, phase angle; STS, Society of Thoracic Surgeons.